

CareGraph

An uncertainty-aware, country-adaptable referral orchestration layer for primary healthcare

World Bank Small AI for Development — Health | Hackathon Working Concept | 3 October 2026

1. Product thesis

CareGraph does not replace WHO guidance or national clinical protocols. It operationalizes them. A frontline worker or patient can type or speak a case; CareGraph retrieves relevant history, asks only decision-relevant clarifying questions, builds a structured case, runs validated triage logic, estimates operational uncertainty, and coordinates the safest executable next action — including peer review, facility selection, referral, transport, handoff and follow-up.

Core objective: minimize time to a safe, executable next action — not maximize confidence in a diagnosis.

2. WHO is the baseline; countries are the operating layer

WHO itself designs SMART Guidelines for localization. WHO states that its Digital Adaptation Kits are generic, software-neutral starting points that governments and implementers must adapt to local policies, procedures, digital tools and health-system context. This is not a workaround around WHO guidance; it is the intended implementation model.

Layer	What it contributes	Examples
Global evidence baseline	Evidence-based recommendations and interoperable representations	WHO SMART Guidelines, DAKs, IITT/ETAT/IMCI, WHO terminology mappings
National adaptation	Country-approved rules and policies	Ministry of Health protocols, referral hierarchy, formularies, national emergency criteria, national terminology/value sets
Subnational context	What is true here and now	District outbreaks, seasonal disease burden, road/transport conditions, facility capability, specialist schedules, bed/service availability
Patient context	What is true for this person	Longitudinal history, medications, allergies, pregnancy, prior encounters, observations, previous referrals
Operational network	What action is actually possible	Facility acceptance, travel time, ambulance/transport, peer availability, escalation chain, counter-referral

Country-specific information is therefore not an optional add-on. It is where a generic clinical recommendation becomes an executable referral decision.

3. The gap CareGraph targets

Do not build another symptom checker. Existing WHO/World Bank ecosystems already cover clinical decision support, digital triage and telehealth. WHO's machine-readable HIV requirements, for example, explicitly call for finding facilities that provide unavailable services, sharing records with referral facilities, and receiving feedback after referral.

CareGraph's differentiation is the agentic orchestration between “this patient may need escalation” and “the patient actually reached appropriate care.”

4. Agentic workflow

1. **Type or speak** — Frontline worker/patient describes the problem in natural language.
2. **Retrieve context** — Pull patient history, medications, allergies, pregnancy, prior encounters/referrals and current observations when authorized.
3. **Build only from evidence** — Extract stated facts. Mark inferred, conflicting and unknown fields separately. Never silently convert missing data into a negative.
4. **Ask the next best question** — Select the minimum clarifying question or measurement most likely to change the next clinical/operational decision.
5. **Call trusted tools** — WHO/national knowledge base, medical terminology/coding, validated risk tools, regional surveillance, facility capability, travel/transport, specialist network.
6. **Estimate decision state** — Separate clinical risk, urgency, evidence completeness, action uncertainty and operational uncertainty.
7. **Run standard triage** — Use deterministic WHO/national logic for red flags and escalation thresholds. If an emergency criterion is already met, stop questioning and escalate.
8. **Choose executable destination** — Match required capability to facilities, current availability/acceptance and realistic travel time — nearest is not necessarily appropriate.
9. **Escalate to humans when needed** — Summarize the case and route it to the smallest appropriate available expertise node with an urgency-dependent response window.
10. **Coordinate handoff** — Request facility acceptance; initiate authorized transport/ambulance workflow; send referral packet; track accepted → departed → arrived.
11. **Close the loop** — Capture receiving-facility disposition/counter-referral and update longitudinal history.

5. CareGraph tool registry

Tool	Authoritative source	Purpose
get_patient_history	FHIR/EMR or demo record	Longitudinal context
get_current_observations	Device/manual entry	Vitals and observations
clinical_knowledge	WHO + country-approved KB	Evidence and approved pathways
medical_code_lookup	ICD/SNOMED/LOINC where permitted	Normalize terms/codes
regional_surveillance	MoH/WHO/public-health feeds	Current local epidemiological context
forecast_risk	Validated model only	Deterioration or pathway-specific risk
run_triage_protocol	Deterministic rules	Urgency / escalation
facility_capability	Facility registry	Services/equipment/specialists
facility_capacity	Facility feed or simulated API	Acceptance/availability
travel_time	Routing/transport data	Realistic time-to-care
expert_network	Credentialed provider directory	Peer/specialist escalation
send_case	WhatsApp/SMS/clinical messaging	Minimal decision packet
request_transfer	Referral workflow	Receiving-facility acceptance
request_transport	Authorized dispatch integration	Ambulance/transport
track_referral	Referral state machine	Closed-loop handoff

6. Uncertainty: what CareGraph should score

Avoid a fake single “AI confidence = 82%.” The product should expose uncertainty that changes action:

- Evidence completeness — are decision-critical observations missing?
- Evidence conflict — do history, observations, guidelines or human opinions disagree?
- Clinical/action uncertainty — is there enough evidence to choose the next safe action even if diagnosis remains uncertain?
- Operational uncertainty — do we know which facility can actually provide the required service now?
- Human acknowledgement — has an authorized clinician/facility accepted the handoff?
- Time sensitivity — is the value of more information lower than the cost of delaying escalation?

7. Country Pack architecture

A deployable CareGraph instance should load a versioned Country Pack rather than hard-code one universal workflow. For a hackathon, demonstrate one fictionalized/clearly labeled country or district pack built from public guidance; a production deployment would require Ministry of Health validation.

- National clinical/referral protocols and permitted local adaptations of WHO guidance
- Referral levels and facility/service taxonomy
- National emergency numbers and transport/escalation workflows
- Drug/formulary and diagnostic availability constraints
- Terminology mappings and required reporting codes
- Language/local terminology and culturally appropriate question phrasing
- Regional surveillance feeds and outbreak alerts
- Facility registry, specialist capabilities, hours and contact pathways
- Consent, privacy, data residency and human-authorization requirements
- Versioning, provenance, effective dates and approval authority for every rule

8. Three-case demo

Case	What happens	What it proves	End state
A. Maternal red flag	Voice case + history + BP. National/WHO rule triggers. Agent stops asking questions.	Protocol authority, urgency, facility capability and transport orchestration.	Accepted referral + simulated transport/handoff
B. Ambiguous child case	Iterative questions reduce uncertainty but not enough. Agent routes concise case to appropriate pediatric peer.	AI knows when not to guess; scarce expert attention is allocated intelligently.	Human-reviewed referral / next action
C. Fever during local outbreak	Patient history + current regional surveillance modify context; protocol selects appropriate local test/care pathway.	Country/subnational context can prevent both under-triage and unnecessary tertiary referral.	Local action + follow-up

9. Hackathon partner software

ElevenLabs — access layer. Multilingual speech-to-text and voice output for frontline workers; preserve original transcript alongside structured extraction.

Bright Data — changing public context layer. Retrieve current public facility/service pages, Ministry/WHO public-health notices and other allow-listed operational sources. Newly retrieved clinical claims must not automatically become decision rules.

Lovable — judge-facing product surface. Build the CHW/patient intake, evidence/uncertainty panel, referral map/network, expert-message simulation and closed-loop status timeline.

LLM/orchestrator — tool selection and communication. Extract observations, choose which approved tool to call, generate concise explanations and case summaries. It must not override deterministic red-flag rules.

Deterministic services — safety-critical logic. Triage rules, thresholds, facility eligibility constraints, state transitions and audit logging.

10. Safety and governance boundaries

- CareGraph is decision support and referral orchestration, not autonomous diagnosis or prescribing.
- Emergency/red-flag rules come from validated WHO/national protocols, not free-form LLM reasoning.
- Every rule/knowledge edge records source, jurisdiction, version, effective date and approval status.

- Patient-specific human advice is attributed to that clinician/case; it is not silently promoted into global medical knowledge.
- High-consequence external actions (e.g., dispatch/transfer) require the authorization appropriate to the local health system.
- Minimize patient data in peer messages; use consent, access control, encryption and auditable handoff states.
- Live web retrieval can inform operational context, but unverified web content cannot become clinical truth.

11. 24-hour MVP

Build one pathway deeply, not a universal medical system. Recommended MVP: maternal urgent referral + one ambiguous pediatric peer-escalation case.

- One multilingual voice intake flow
- One small versioned WHO/national-style protocol pack
- One structured patient history record
- One evidence/unknown/conflict state representation
- One iterative-question selector
- One deterministic triage engine
- 10–15 synthetic facilities with capabilities/availability/travel time
- 5–8 synthetic expert nodes with specialty/availability
- WhatsApp-style peer escalation simulation
- Referral state machine: created → accepted → transport → arrived → closed
- One Bright Data verification/current-context action
- Audit trail showing which tool/rule/evidence produced each action

12. Pitch

“WHO tells us what good care should look like. Countries know how their health systems actually work. CareGraph connects the two — asking only what matters, recognizing uncertainty, and coordinating the right patient to the right care at the right urgency until the handoff is complete.”

13. Evidence basis

WHO SMART Guidelines: WHO states that global recommendations must be interpreted and adapted to local policies, procedures and digital tools; SMART content is designed for localization and interoperability.

<https://www.who.int/teams/digital-health-and-innovation/smart-guidelines>

WHO SMART DAK country adaptation: DAKs provide generic software-neutral requirements that must be adapted to local needs. <https://smart.who.int/dak-smbp/adapting.html>

WHO SMART HIV referral requirements: Requirements include locating facilities that can provide unavailable services, sharing health records, contact information and receiving feedback after referral.

<https://smart.who.int/hiv/Requirements-HIV.I.Referral.html>

WHO Digital Adaptation Kit for TB: DAKs include workflows, core data elements mapped to standard terminology, decision support, indicators and functional requirements. <https://www.who.int/publications/i/item/9789240086616>

WHO FHIR-based SMART Guidelines: WHO describes country-specific, offline-capable and interoperable digital applications built from standard components.

<https://www.who.int/teams/digital-health-and-innovation/smart-guidelines/fhir-based-smart-guidelines>